

citemsp: Citation Locators for L^AT_EX

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Abstract

We present `citemsp`, a L^AT_EX package that adds per-key locators to numeric citation labels, rendered as compact superscript/subscript pairs¹. Working on top of either `biblatex` or `natbib`, the package provides a slash-delimited syntax with ten built-in prefix types — covering sections, equations, theorems, figures, and more — and a user-extensible registry for domain-specific locators.

1 Syntax and Usage

The `\citemsp` command accepts a comma-separated list of citation keys with optional locators separated by forward slashes. By default, the first locator slot renders as § (section) and the second as ¶ (paragraph). A single-letter prefix changes the locator type. With no locators, `\citemsp` behaves identically to `\cite`, including range compression when using `numeric-comp` or `sort&compress`.

<code>\citemsp{key}</code>	→	$[1]$
<code>\citemsp{key/section}</code>	→	$[1]^{§6.2}$
<code>\citemsp{key/section/paragraph}</code>	→	$[1]^{§6.2}_{¶3}$
<code>\citemsp{key//paragraph}</code>	→	$[1]_{¶3}$
<code>\citemsp{key/section/e4.3}</code>	→	$[1]^{§6.2}_{eq.4.3}$

Currently there exist ten built-in prefixes, which are listed in Table 1. Each prefix can appear in either the superscript or subscript slot. Users can register additional prefixes with `\citemspprefix{letter}{label}`.

2 Examples

¹The package is available at <https://github.com/ApostolosTsabodimos/citemsp-Latex-package> and on CTAN.

Prefix	Locator	Render	In super slot	In sub slot	Both slots
A	Appendix	App.	$[1^{\text{App.2}}]$	$[1^{\text{App.2}}]$	$[1^{\S 3.2}_{\text{App.2}}]$
C	Corollary	Cor.	$[1^{\text{Cor.3}}]$	$[1^{\text{Cor.3}}]$	$[1^{\S 4.1}_{\text{Cor.3}}]$
d	Definition	$\triangle =$	$[1^{\triangle=5}]$	$[1^{\triangle=5}]$	$[1^{\S 3.2}_{\triangle=5}]$
e	Equation	eq.	$[1^{\text{eq.4.3}}]$	$[1^{\text{eq.4.3}}]$	$[1^{\S 6.2}_{\text{eq.4.3}}]$
f	Figure	fig.	$[1^{\text{fig.3}}]$	$[1^{\text{fig.3}}]$	$[1^{\S 6.1}_{\text{fig.3}}]$
L	Lemma	Lem.	$[1^{\text{Lem.1}}]$	$[1^{\text{Lem.1}}]$	$[1^{\S 3.2}_{\text{Lem.1}}]$
n	Footnote	†	$[2^{\text{†7}}]$	$[2^{\text{†7}}]$	$[2^{\S 1}_{\text{†7}}]$
p	Page	pp.	$[1^{\text{pp.42}}]$	$[1^{\text{pp.42}}]$	$[1^{\S 3.2}_{\text{pp.42}}]$
t	Table	■+	$[3^{\text{■+2}}]$	$[3^{\text{■+2}}]$	$[3^{\S 8.4}_{\text{■+2}}]$
T	Theorem	Th.	$[1^{\text{Th.4.2}}]$	$[1^{\text{Th.4.2}}]$	$[1^{\S 6.1}_{\text{Th.4.2}}]$

Table 1: Built-in prefix registry.

2.1 Basic Usage

The Schwarzschild metric~\citemsp{wald1984/6.1/2} describes spherically symmetric vacuum solutions.

Output: The Schwarzschild metric $[1^{\S 6.1}_{\text{†2}}]$ describes spherically symmetric vacuum solutions.

Multiple sources with independent locators:

Black hole thermodynamics~\citemsp{hawking1974/1/2, wald1984/12.5/1} establishes a deep connection between gravity and quantum mechanics.

Output: Black hole thermodynamics $[2^{\S 1}_{\text{†2}}, 1^{\S 12.5}_{\text{†1}}]$ establishes a deep connection between gravity and quantum mechanics.

Locator entries can be freely mixed:

General covariance~\citemsp{einstein1915, wald1984/4, hawking1974/1/2} forms the foundation of general relativity.

Output: General covariance $[4, 1^{\S 4}, 2^{\S 1}_{\text{†2}}]$ forms the foundation of general relativity.

2.2 Similarity to \cite

With no locators, \citemsp passes keys through the backend’s native \cite, producing identical output:

\cite{wald1984} $\longrightarrow [1]$ \citemsp{wald1984} $\longrightarrow [1]$

All plain (5 consecutive): $[1\text{--}5]$

All plain (non-consecutive): $[1, 4\text{--}8]$

Range, then locator: $[1, 4, 5, 2^{\S 3.2}]$

Locator, then range: $[4^{\S 3.2}, 1\text{--}3, 5]$

Range, locator, range: $[1, 4, 5, 2^{\text{eq.2.1}}, 3, 6, 7]$

Adjacent locators only: $[1^{\S 3.1}, 2^{\text{eq.2.1}}, 3^{\text{pp.42}}]$

2.3 Mixed Types and Custom Prefixes

All built-in locator types can be freely mixed in a single call:

```
See~\citemsp{wald1984/T6.2, einstein1915/3.2/d5,  
mtw1973/p520, hawking1974/1/e4.3} for further details.
```

Output: See [$1^{\text{Th.6.2}}$, $4_{\Delta=5}^{\S 3.2}$, $3^{\text{pp.520}}$, $2_{\text{eq.4.3}}^{\S 1}$] for further details.

Users can register domain-specific prefixes with `\citemspprefix`. For example, a cosmology paper might define prefixes for window functions and sky patches:

```
\citemspprefix{w}{w.}  
\citemspprefix{S}{\ensuremath{\Omega}}  
The top-hat window~\citemsp{wald1984/4.2/w3} smooths  
the power spectrum over sky patch~\citemsp{mtw1973/S12}.
```

Output: The top-hat window [$1_{w,3}^{\S 4.2}$] smooths the power spectrum over sky patch [$3^{\Omega 12}$].

2.4 Paragraph example

The following paragraph illustrates how `\citemsp` reads in running text, mixing plain citations, section locators, and prefix types:

```
The Schwarzschild solution~\citemsp{schwarzschild1916/1/3}  
was found shortly after the field  
equations~\citemsp{einstein1915} were published. A modern  
treatment can be found in standard  
textbooks~\citemsp{wald1984/6.1/2, mtw1973/p520}, where the  
derivation proceeds via the Birkhoff  
theorem~\citemsp{wald1984/T6.2}. The singularity  
theorems~\citemsp{penrose1965, hawking1974/1/e4.3} later  
showed that gravitational collapse is generic. For further  
context see~\citemsp{kerr1963, chandrasekhar1931}.
```

Output: The Schwarzschild solution [$5_{\S 1}^{\S 1}$] was found shortly after the field equations [4] were published. A modern treatment can be found in standard textbooks [$1_{\S 2}^{\S 6.1}$, $3^{\text{pp.520}}$], where the derivation proceeds via the Birkhoff theorem [$1^{\text{Th.6.2}}$]. The singularity theorems [$6, 2_{\text{eq.4.3}}^{\S 1}$] later showed that gravitational collapse is generic. For further context see [$7, 8$].

3 Loading and Configuration

With biblatex

```
\usepackage[style=numeric-comp, backend=biber]{biblatex}  
\addbibresource{refs.bib}  
\usepackage{citemsp}
```

Use `numeric-comp` (or `numeric`) as the style. Range compression of consecutive plain entries requires `numeric-comp`.

Configuration

The locator glyph size is controlled by `\citersp scale` (default 0.35):

Default (0.35): $[1^{§6.1}_{eq.4.3}]$ Smaller (0.25): $[1^{§6.1}_{eq.4.3}]$ Larger (0.50): $[1^{§6.1}_{eq.4.3}]$

To adjust the locator glyph size, redefine `\citersp scale` after loading:

```
\usepackage{citemsp}
\renewcommand{\citersp scale}{0.4} % default is 0.35
```

4 Relation to Standard Locators

The standard `biblatex` provides locators via its postnote mechanism. The `\citemsp` package replaces that mechanism with a more compact notation. We illustrate the differences below.

Side-by-side source comparison

Consider citing three sources, each with a different locator type:

```
% Standard biblatex (71 characters):
\autocites[$6.1, ¶2]{wald1984}[pp.\,520]{mtw1973}
  [$1, eq.\,4.3]{hawking1974}

% citemsp equivalent (53 characters):
\citemsp{wald1984/6.1/2, mtw1973/p520,
  hawking1974/1/e4.3}
```

Rendered output:

biblatex: $[1, §6.1, ¶2, 3, pp. 520, 2, §1, eq. 4.3]$

citemsp: $[1^{§6.1}_{¶2}, 3^{pp. 520}, 2^{§1}_{eq. 4.3}]$

The `citemsp` version is 25% shorter in source characters and uses a flat, comma-separated list rather than nested optional arguments. Locators are rendered as superscript/subscript pairs attached to each citation number, rather than appended as flat text inside the brackets.

Scaling to dense citations

The difference becomes pronounced in review paragraphs with many locator-bearing citations. Consider six sources:

```
% Standard biblatex (6 sources, 185 characters):
\autocites[$6.1, ¶2]{wald1984}[pp.\,520]{mtw1973}
  [$1, eq.\,4.3]{hawking1974}[$3]{penrose1965}
  [$2]{kerr1963}[ch.\,1]{chandrasekhar1931}

% citemsp equivalent (6 sources, 107 characters):
\citemsp{wald1984/6.1/2, mtw1973/p520,
  hawking1974/1/e4.3, penrose1965/3,
  kerr1963/2, chandrasekhar1931/1}
```

Rendered output:

biblatex: [1, §6.1, ¶2, 3, pp. 520, 2, §1, eq. 4.3, 6, §3, 7, §2, 8, ch. 1]

citemsp: [^{§6.1}_{¶2}, ^{pp. 520}3, ²_{eq. 4.3}, ^{§3}6, ⁷_{§2}, ⁸_{ch. 1}]

At six entries, the `\autocites` source is 78 characters longer (a 42% reduction) and produces a bracket sequence that occupies significant horizontal space in the running text. The `citemsp` version remains a single bracket with compact locator annotations.

Summary of advantages

Three properties distinguish `\citemsp`: (1) a compact, uniform syntax that stays readable as entries accumulate; (2) typed locator prefixes that visually distinguish sections, equations, pages, theorems, and other reference types; and (3) superscript/subscript rendering that keeps locator information close to each citation number rather than appended as flat text inside the brackets.

5 Conventions

- **Section numbering.** Use the source’s native section numbering format (e.g., 6.1 for “Section 6.1”).
- **Paragraph counting.** Paragraphs are counted manually from the start of the section. This is inherently edition-dependent and best suited for stable references such as textbooks. Since paragraphs are not explicitly numbered in published sources, these locators may be ambiguous across editions or between readers. When using ¶ locators, consider providing additional context in the surrounding text so the reader can locate the passage independently.
- **Custom prefixes.** Register new locator types with `\citemspprefix{letter}{label}`. The letter must be a single character not already in use. Custom and built-in prefixes can be freely mixed in the same `\citemsp` call.

References

- [1] Robert M. Wald. *General Relativity*. University of Chicago Press, 1984.
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- [3] Charles W. Misner, Kip S. Thorne, and John Archibald Wheeler. *Gravitation*. W. H. Freeman, 1973.
- [4] Albert Einstein. “Die Feldgleichungen der Gravitation”. In: *Sitzungsberichte der Preussischen Akademie der Wissenschaften* (1915), pp. 844–847.
- [5] Karl Schwarzschild. “Über das Gravitationsfeld eines Massenpunktes nach der Einsteinschen Theorie”. In: *Sitzungsberichte der Preussischen Akademie der Wissenschaften* (1916), pp. 189–196.
- [6] Roger Penrose. “Gravitational Collapse and Space-Time Singularities”. In: *Physical Review Letters* 14 (1965), pp. 57–59.
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- [8] Subrahmanyan Chandrasekhar. “The Maximum Mass of Ideal White Dwarfs”. In: *The Astrophysical Journal* 74 (1931), pp. 81–82.